MSC ARTIFICIAL INTELLIGENCE DISSERTATION

Learning to Select Actions for Complex Hand Tasks

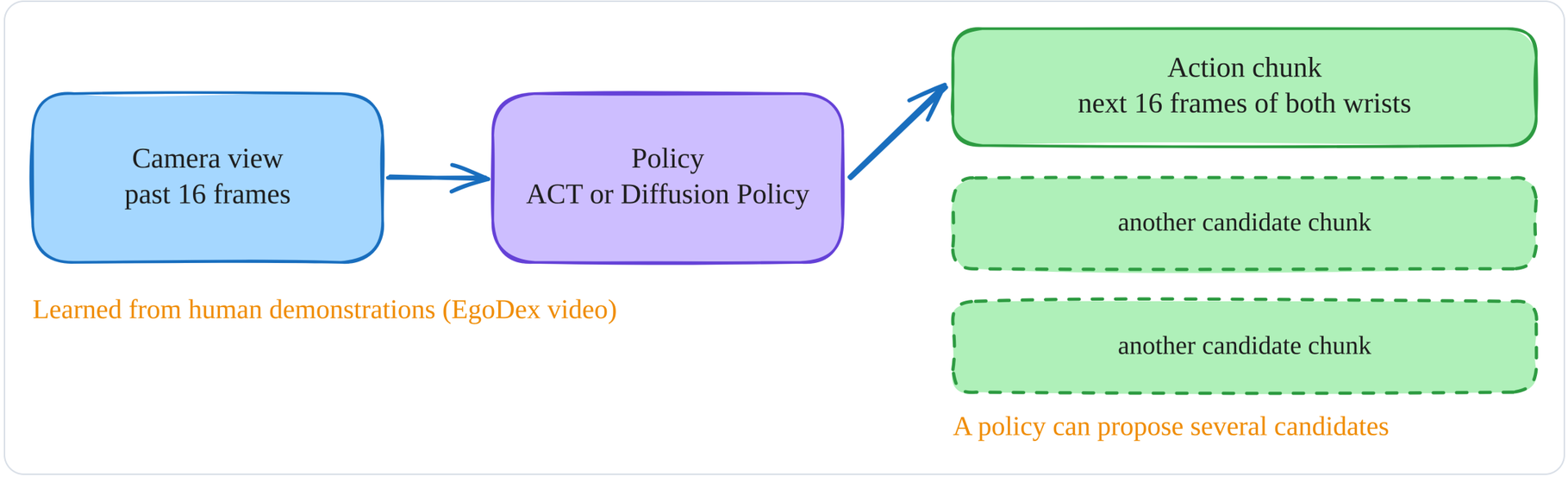
A comparative study of visuomotor policies and latent world models

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THE FIELD IN ONE PICTURE

A policy turns what a robot sees into the motion it makes next

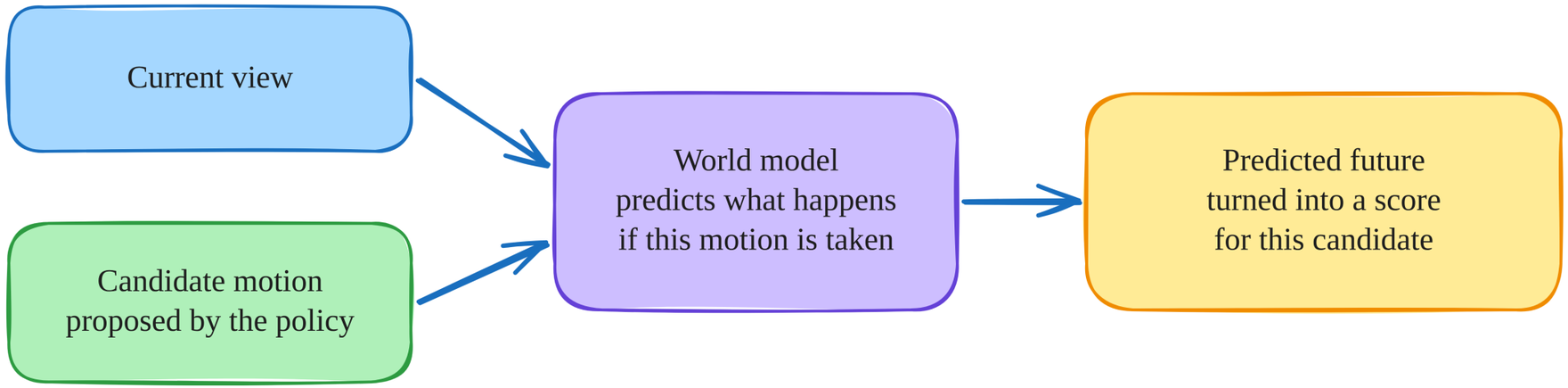
It is trained from human demonstrations. Here the demonstrations are egocentric hand video, and the output is a short chunk of wrist motion.



THE SECOND BUILDING BLOCK

A world model predicts what happens next if a motion is taken

It does not decide what to do. It answers a what-if question, so a motion can be judged before anything is executed.

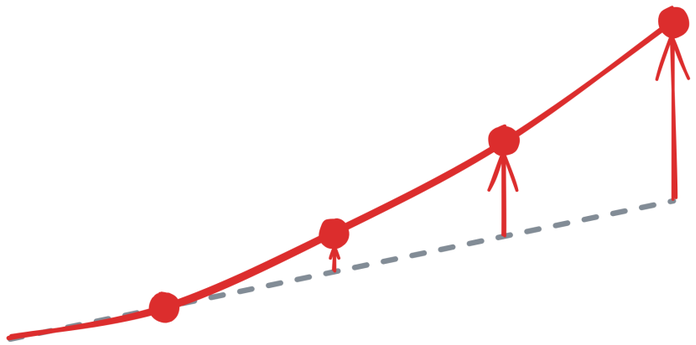


It never generates motion here. It only predicts what a motion would lead to.

WHY THIS MATTERS

Small errors compound, and a single guess is brittle

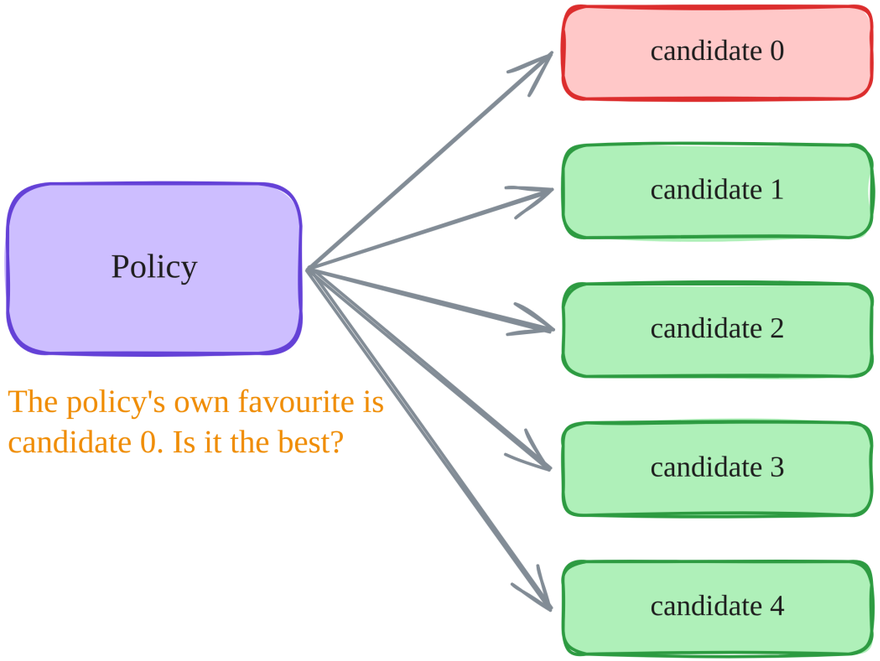
Small errors compound



expert motion policy rollout

A slightly wrong move early drifts further as the sequence unfolds

One guess is brittle



The policy's own favourite is candidate 0. Is it the best?

WHERE THE QUESTION COMES FROM

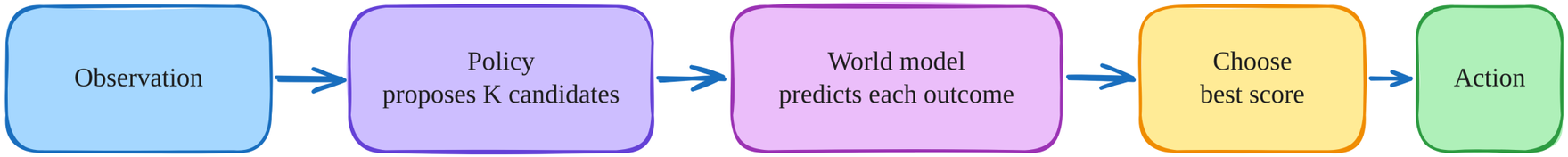
Act directly, or look ahead and then choose

VLA: one network, one answer



Direct mapping to an action, with no separate look-ahead step

Policy + world model (LeCun-style): look ahead, then choose

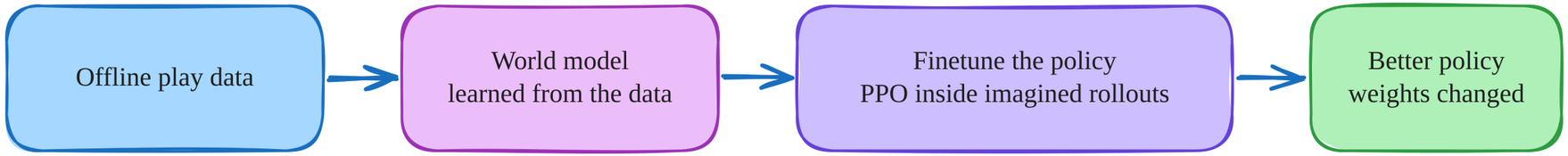


Our research question lives here: the world model is a frozen judge, not a generator

THE CLOSEST PRIOR WORK

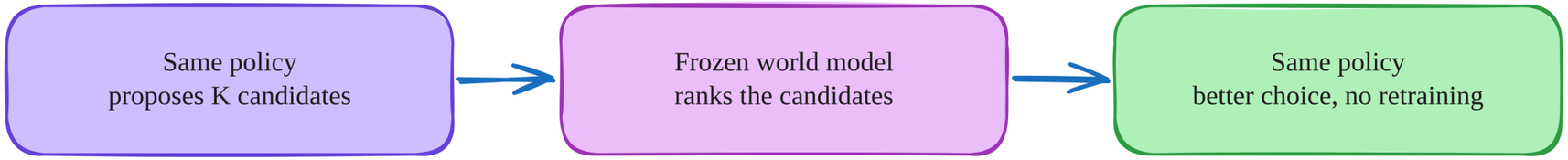
DiWA uses a world model to retrain the policy. This study uses it only to choose.

DiWA (Chandra et al., 2025): change the policy



Evaluated on CALVIN: the policy gets better because it was retrained

This study: leave the policy alone, only choose



Open question: can a frozen world model help just by choosing?

FROM THE GAP TO THE QUESTION

Can a frozen world model help just by choosing?

RESEARCH QUESTION

Can a frozen world model, acting only as the final decision stage over candidates a policy already proposes, improve the selected motion chunk?

Cheap

The policy is never retrained. Selection can sit on top of a policy that already exists.

Clean

The candidates are fixed, so any change comes from the world model's judgement and nothing else.

Safe

The world model never generates motion. It only ranks motions the policy already proposed.

WHAT THE STUDY DOES

Three objectives: build, compare, diagnose

Build

A controlled offline benchmark that separates the policy's preference from the world model's.

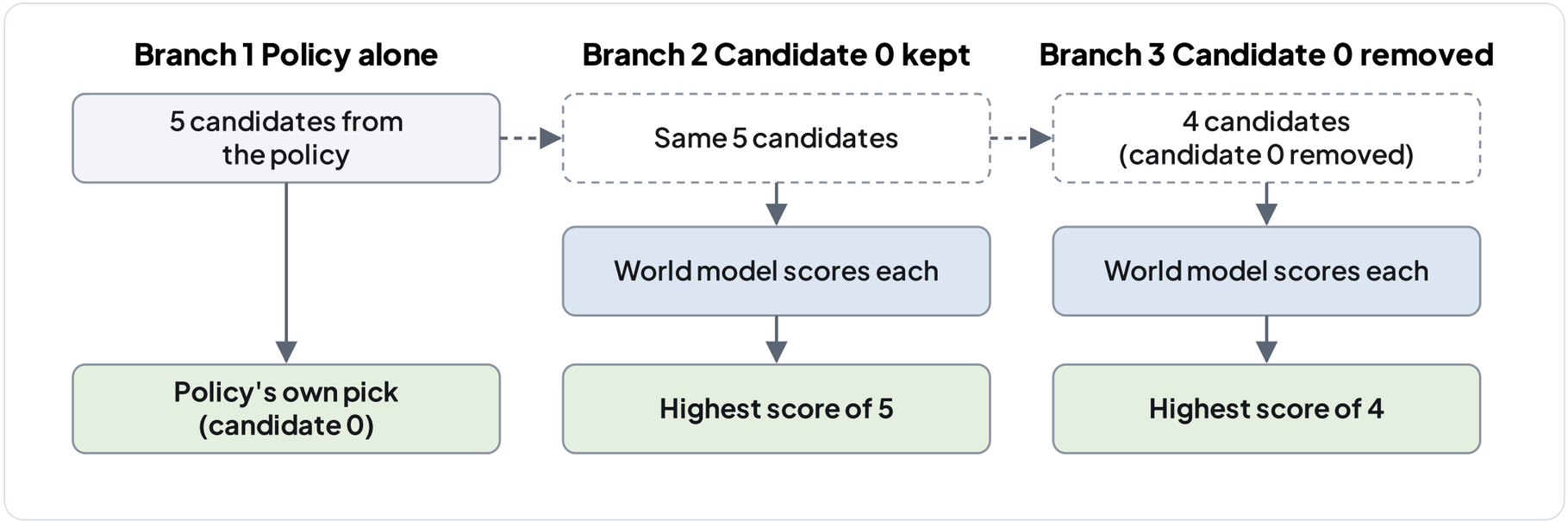
Compare

Five world-model families and two policies, under matched local training exposure.

Diagnose

When selection helps, and a likely explanation for each case where it does not.

One pool of candidates, three ways to choose



7,607	16 to 16	18-D	K = 5	3 seeds
HELD-OUT WINDOWS	FRAMES IN, FRAMES OUT	BIMANUAL WRIST MOTION	CANDIDATES PER WINDOW	POOLED RESULTS

Data is EgoDex egocentric video. Error is measured against the expert motion as translation (metres, per axis) and rotation (degrees). Branch 1 is the baseline everything is compared with.

Where the candidates come from changes what the result means

Diffusion Policy

5 noise seeds → 5 distinct motions → World model ranks them

Independent pool, clean measurement

ACT

5 latent samples → 1 identical motion (posterior collapse) → CEMP search scored by the world model

Scorer shapes its own pool before selecting

WHAT HAPPENED TO ACT

Four retraining attempts changed the KL weight, annealing, observation dropout and free bits. The latent was still ignored, and 7 of 12 checkpoints collapsed across seeds.

WHAT THAT MEANS

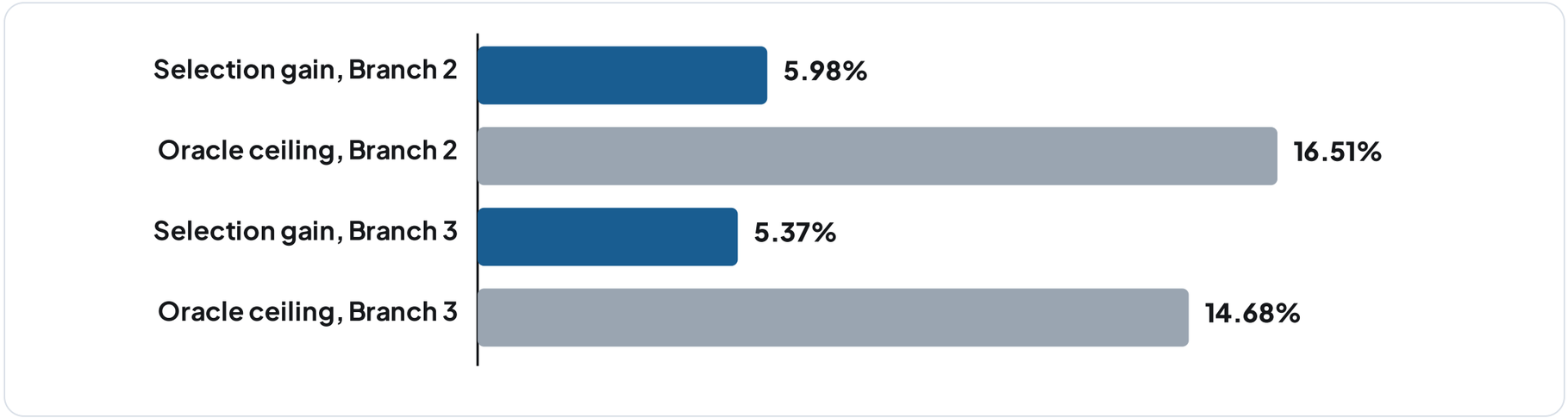
ACT results measure the world model and planner together. Diffusion Policy has no such coupling, so it carries the main claim.

Five world-model families, plus two controls

MODEL	WHAT IT IS	ROLE IN THIS STUDY
V-JEPA2-AC	Frozen video backbone pretrained on over a million hours, with an action-conditioned head trained locally	Main candidate scorer
DexWM	Dexterous-manipulation world model on a DINOv2 encoder	Trained at a fixed budget, one seed
LeWM	Compact encoder and predictor trained from scratch	Two scorer configurations
DINO-WM, JEPA-WM	Official checkpoints with a 7-D action interface	Validation pilot through an adapter
No-AC, PointWorld proxy	Same backbone with the action input removed or weakened	Controls for the mechanism

Every locally trained model receives the same exposure of 100,000 items. Externally pretrained backbones still carry training that budget does not offset, so results are read model by model.

Ranking Diffusion Policy candidates with V-JEPA2-AC cuts translation error by 5.98%



-0.96° ROTATION ERROR	**about 36%** OF ORACLE HEADROOM	**49.9 / 23.2 / 27.0** WIN / TIE / LOSS, %	**0.00%** WITHOUT ACTION INPUT

Pooled across three training seeds, 95% CI [5.43, 6.47]. No retraining. The oracle ceiling is what a perfect ground-truth selector could remove from the same five candidates. Translation error is root-mean-square per axis.

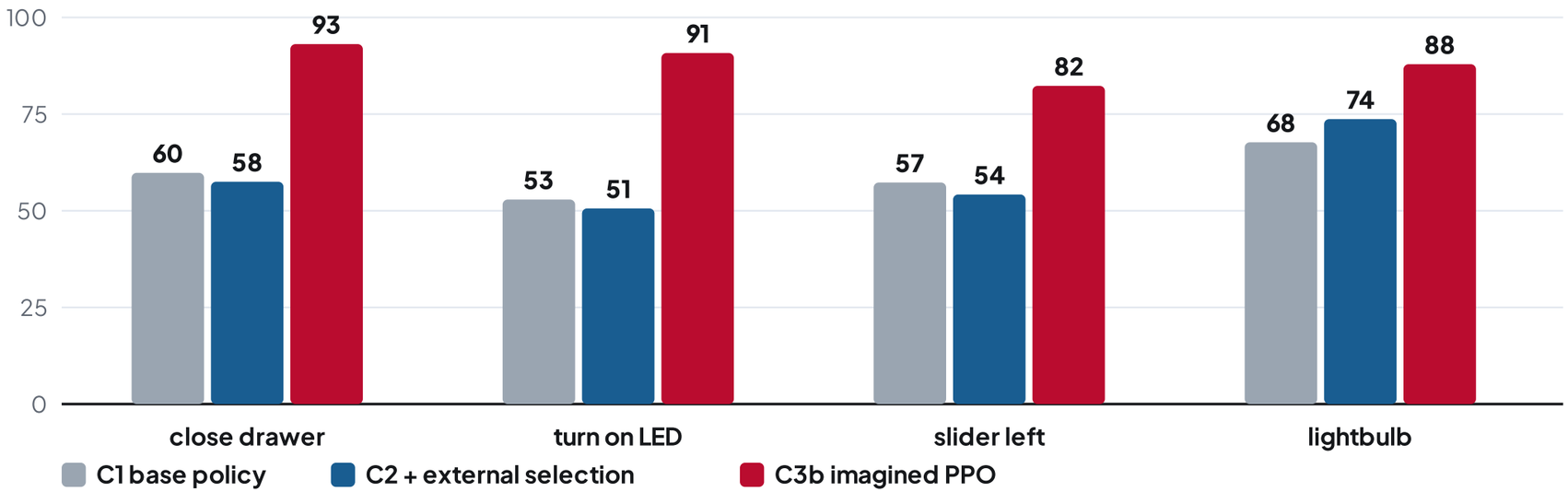
The other routes did not reproduce it, and one result needs care

LEWM: THE LARGEST NUMBER, NOT THE BEST EVIDENCE

LeWM shows a 10.1% translation reduction, but its score does not predict which candidate is better (rank correlation −0.51 on the full test set). Disabling its dynamics model gives almost the same gain. It is reported as unvalidated and held out of the headline.

ROUTE	OUTCOME	LIKELY EXPLANATION
DexWM	+0.57% worse one seed	Predictor trained at a fixed budget, published checkpoint unavailable
DINO-WM, JEPA-WM	Latent score only	Adapter squeezes 18-D actions into a 7-D interface
PointWorld proxy	+0.27% not significant	Weakened action input, a control by design
ACT with CEMP	−0.58% small	Scorer shapes its own pool, read as a unit

Finetuning inside a world model beats selecting with one



SELECTION (C2 VS C1)

Moves tasks by −3.1 to +6.1 points, with overlapping intervals.

FINETUNING (C3B VS C2)

Adds 14 to 40 points on every task, about an order of magnitude more.

Task success (%), four CALVIN tasks, DiWA's own checkpoints, pooled across three seeds.

Selection helped when three conditions held together

1 Diverse pool

There must be real alternatives to choose between. ACT's collapsed pool offered none until search restored it.

2 Action information

The scorer must see the action. Removing it gave exactly 0.00%.

3 In-support candidates

Candidates must lie within what the scorer was trained on. Adapter pilots improved a latent score and not the trajectory.

WHY FINETUNING WINS

DiWA's imagined-PPO finetuning meets all three conditions by construction, because its reward signal is what the policy is trained against. External selection has no such guarantee.

This is a pattern supported by the tested configurations, not a general law.

What this does and does not show

LIMITS

- Offline trajectory error, not robot task success
- One dataset and one positive pairing
- Matched exposure, not equal pretraining
- LeWM's gain is unexplained
- Selection worsens the easiest fifth of windows

NEXT STEPS

- Closed-loop execution, starting with CALVIN-style tasks
- A second, independent world model as selector over a finetuned policy
- Native 18-D finetuning of DINO-WM and JEPA-WM
- DexWM at its published scale
- A diversity mechanism for ACT

Contribution in one sentence. A controlled offline benchmark and protocol showing a frozen world model can improve which of a policy's own proposed motions is chosen, for one tested pairing, without retraining and without any claim yet about real-robot success.

THANK YOU FOR LISTENING

Questions

Backup slides follow, covering the AI-use workflow, how each method was chosen, LeWM's gate, seeds, pool size, and where selection fails.

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How AI was used in this work

NotebookLM

Queried paper sets for related work and citations, then each source was skimmed manually.

Claude skills

Query, draft and polish workflows, with my own review and edits.

Gemini

Paper discovery and critical review of chapters against a rubric.

Coding agents

Wrote most of the code under my specification, then tested and audited.

WHERE THE DECISIONS SAT

Research question, benchmark design, which methods to adopt, what counted as a valid result, and every claim in the dissertation.

HOW CORRECTNESS WAS CHECKED

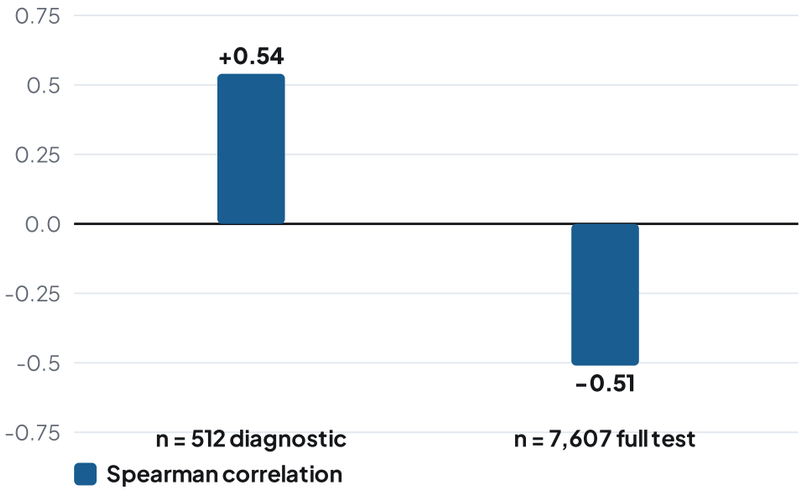
Separate audit passes and re-runs found real defects, including a DDIM sampler indexing error, a training-exposure undershoot, and a DiWA configuration mistake. All were fixed before reporting.

Used most in the literature review and methodology, for example to find established reference methods for handling ACT and LeWM without inventing new verification tests.

Which published work anchored each method choice

CHOICE	ANCHOR	STATUS
Policy-guided candidate search (CEMP)	Cross-entropy method, initialised from the policy output, inspired by TD-MPC2	Adapted, not a published recipe
Posterior-collapse fixes for ACT	KL annealing (Fu et al.), free bits (Kingma et al.), observation dropout	All tried, all failed
Diffusion sampling	DDIM (Song et al.), DP-T architecture (Chi et al.)	Implemented, sampler bug fixed
Action-conditioned scorer	V-JEPA2-AC (Assran et al.)	Head trained locally
Finetuning comparison	DiWA (Chandra et al.) on CALVIN, own checkpoints	Reproduced, secondary study
Pool size K = 5	No literature anchor	Compute choice, ranking stable K = 2 to 20
Matched exposure 100,000 items	Compute ceiling, loss still falling	Not a convergence claim

LeWM passes its ranking gate on 512 windows and fails it on 7,607



WHAT IS AND IS NOT ESTABLISHED

- The gain is real, −10.1% translation, intervals exclude zero
- The reported configuration never invokes LeWM's learned dynamics
- Its score does not predict which candidate is better at full scale
- Reweighting terms to pass the gate was rejected as overfitting the gate

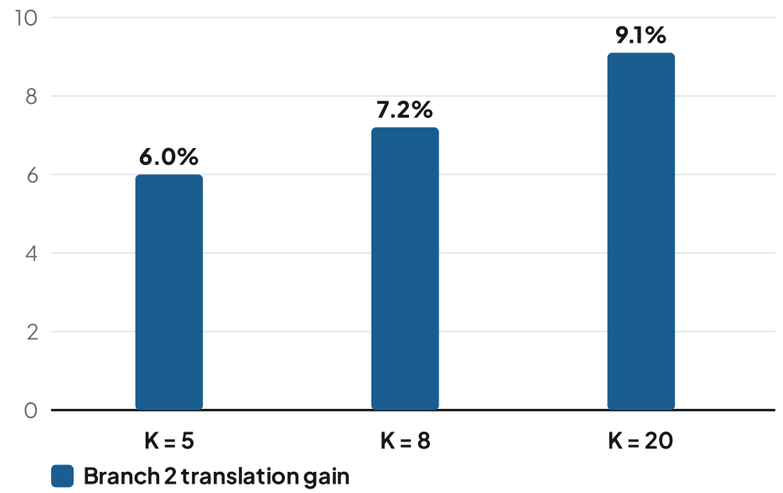
Spearman correlation between score margin and true translation improvement. The 512-window figure was +0.537, the full test set gave −0.51.

The result holds at every seed, and the controls behave as designed

	SEED 42	SEED 43	SEED 44
Baseline translation error	0.1142	0.1178	0.1191
V-JEPA2-AC, Branch 2	-6.41%	-6.09%	-5.46%
V-JEPA2 without action input	0.00%	0.00%	0.00%
PointWorld proxy	+0.50%	+0.23%	+0.07%
LeWM rollout (unvalidated)	-10.10%	-10.06%	-10.12%

Three seeds are a basic robustness check, not proof of stability. DexWM is single-seed by a cost decision of roughly 121 extra GPU-hours.

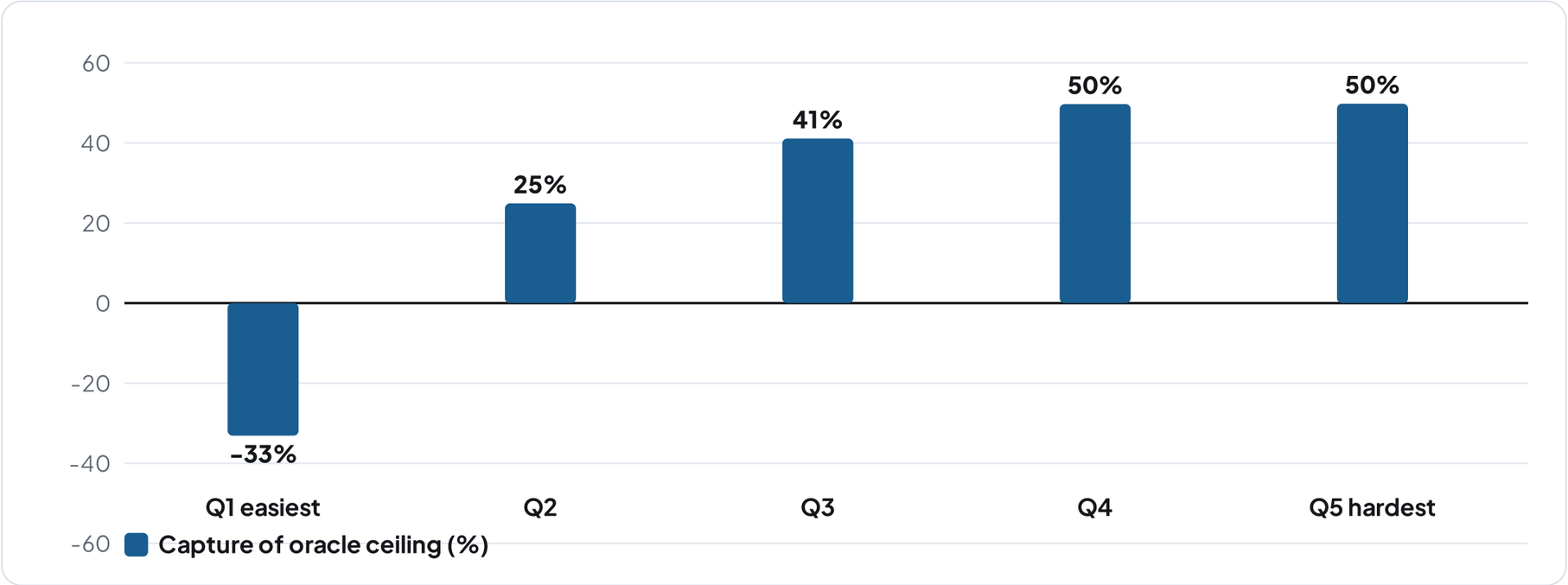
More candidates buy more headroom, not better selection



WHAT THE SWEEP SHOWS

- Scorer ranking identical at every K from 2 to 20
- Gain grows steadily with K and does not plateau
- Capture of the oracle ceiling stays near 40%
- K = 5 was a compute choice, with no literature anchor

On the easiest fifth of windows, selection makes things worse



Windows sorted by baseline error, K = 8, Branch 2. Translation error rises 5.81% in Q1. Confidence gating does not help, because the score gap has a correlation of -0.002 with baseline error. A state-change-targeted windowing removes most of the reversal.

Four retraining attempts, the same collapse

RUN	KL WEIGHT	CYCLICAL KL	OBSERVATION DROPOUT	FREE BITS	PAIRWISE DIFFERENCE
v1	10	no	0%	no	indistinguishable from zero
v2	100	no	0%	no	indistinguishable from zero
v3	100	yes	30%	no	indistinguishable from zero
v4	100	yes	0%	0.5 nats/dim	indistinguishable from zero

Collapse is seed-dependent. Under the default configuration 1 of 3 seeds collapsed, and 7 of 12 checkpoints collapsed across all configurations. EgoDex observations are informative enough that the decoder finds a loss minimum that ignores the latent. A repair would need an architectural change such as discrete latents or a diversity head.

How results were measured

ERROR METRICS

- Translation, root-mean-square per axis, not Euclidean distance
- Rotation, mean geodesic angle in degrees
- Both averaged over the full 16-step chunk
- Change reported relative to candidate 0

STATISTICS

- 95% episode-clustered bootstrap, 2,000 resamples, pooled over seeds
- Oracle ceiling, best candidate in the same pool
- Ranking-validity gate, per-window Spearman correlation
- Win, tie, loss against candidate 0

Test windows are fixed, candidate pools frozen, future-goal access blocked, and scorer support restricted to training data.

What is disclosed about test-set use

- The Branch 2 margin threshold's final value came from a disjoint validation split.
- An earlier, superseded calibration touched the test set directly. This is stated as a named limitation and is reassuring, not a full defence.
- The LeWM gate was tuned on a small diagnostic split and failed to generalise to the full test set. It was reported, not re-tuned.
- Earlier stronger-looking results relying on shortcuts, transductive support or oracle access were corrected rather than reported.

Objective achievement

OBJ.	STATUS	BASIS
1	Fully	Three-branch benchmark implemented as designed and used for every family reported
2	Partly	Local exposure matched, externally pretrained backbones carry an advantage not equalised
3	Within tested configurations	Whether and when selection helps determined for every combination, one positive, a likely explanation for each other

Objectives 4 and 5 are in the Conclusion chapter's objective table. Every non-positive-route explanation is the study's own interpretation, not independently proven.